Challenges of Outlier Detection

- Modeling normal objects and outliers effectively
  - The border between data normality and abnormality (outliers) is often not clear-cut
- Application-specific outlier detection
  - The relationship among objects highly depends on applications
- Handling noise in outlier detection
  - Noise often unavoidably exists in data collected in many applications
  - Low data quality and the presence of noise bring a huge challenge to outlier detection
- Interpretability: a user may want to not only detect outliers, but also understand why the detected objects are outliers

Outlier Detection Methods: An Overview

- Supervised, Semi-Supervised, and Unsupervised Methods
- Statistical methods, proximity-based methods, and reconstruction-based methods

Supervised, Semi-Supervised, and Unsupervised Methods

- Supervised methods model data normality and abnormality
  - Imbalance data sets
  - Avoid many false positives in outlier detection
- Unsupervised outlier detection methods make an implicit assumption: the normal objects are somewhat “clustered”
- Semi-supervised methods: although obtaining some labeled examples is feasible, the number of such labeled examples is often small

Statistical Methods, Proximity-based Methods, and Reconstruction-based Methods

- Statistical methods (also known as model-based methods) make assumptions of data normality
  - Example: detecting outliers using a statistical (Gaussian) model
- Proximity-based methods assume that an object is an outlier if the nearest neighbors of the object are far away in feature space, that is, the proximity of the object to its neighbors significantly deviates from the proximity of most of the other objects to their neighbors in the same data set
- Reconstruction-based methods: matrix-factorization based methods and pattern-based compression methods
  - The normal data samples often share certain similarities, they can often be represented in a more succinct way, compared with their original representation
  - With the succinct representation, we can well reconstruct the original representation of the normal samples
  - For samples which cannot be well reconstructed by such alternative, succinct representation, we flag them as outliers